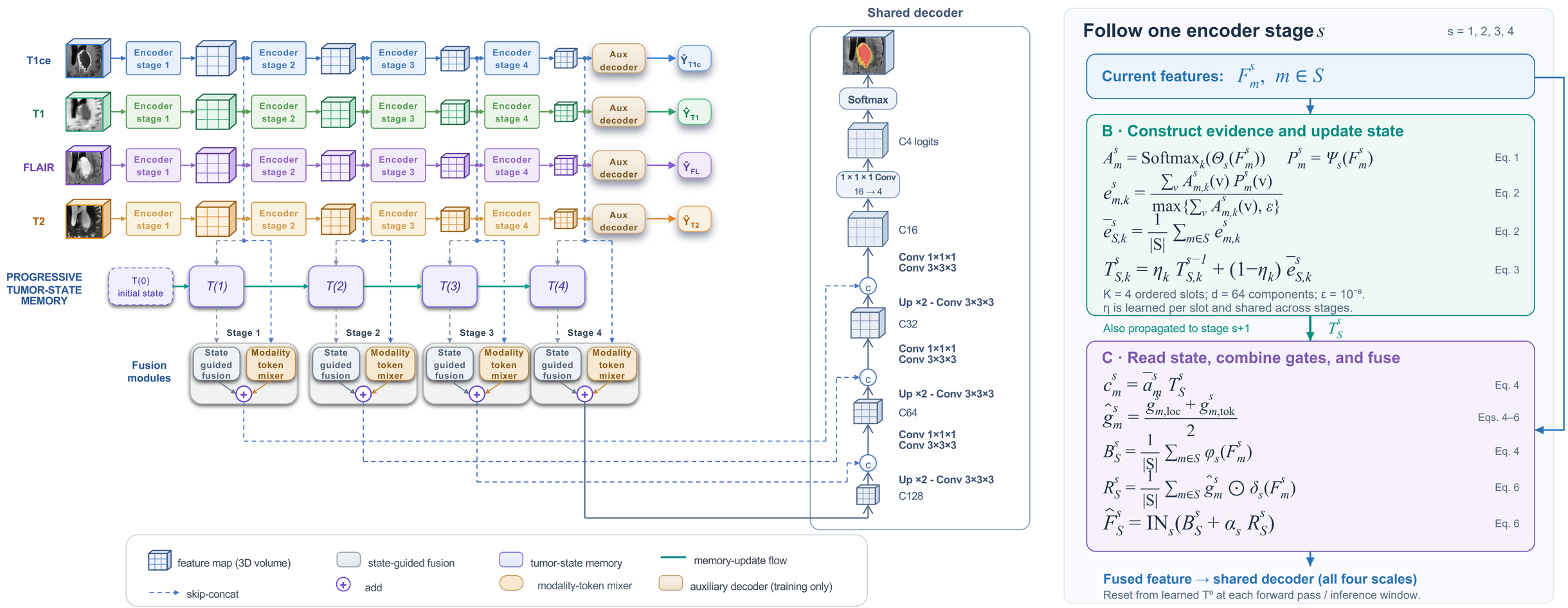
PMTM | Figures 1–3, connected in one view

Original artwork retained; callouts explain how state construction and fusion fit into the network.

A Figure 1 · Network overview



Follow one encoder stage s

s = 1, 2, 3, 4

Current features: $F_m^s,\ m \in S$

B · Construct evidence and update state

$$A_m^s = \mathrm{Softmax}_k(\Theta_s(F_m^s)) \qquad P_m^s = \Psi_s(F_m^s) \quad \text{Eq. 1}$$

$$e_{m,k}^s = \frac{\sum_v A_{m,k}^s(v)\, P_m^s(v)}{\max\{\sum_v A_{m,k}^s(v),\ \varepsilon\}} \quad \text{Eq. 2}$$

$$\bar{e}_{S,k}^s = \frac{1}{|S|}\sum_{m\in S} e_{m,k}^s \quad \text{Eq. 2}$$

$$T_{S,k}^s = \eta_k\, T_{S,k}^{s-1} + (1-\eta_k)\, \bar{e}_{S,k}^s \quad \text{Eq. 3}$$

K = 4 ordered slots; d = 64 components; ε = 10^{-6}.
η is learned per slot and shared across stages.

Also propagated to stage s+1 → T_S^s

C · Read state, combine gates, and fuse

$$c_m^s = \bar{a}_m^s\, T_S^s \quad \text{Eq. 4}$$

$$\hat{g}_m^s = \frac{g_{m,\mathrm{loc}}^s + g_{m,\mathrm{tok}}^s}{2} \quad \text{Eqs. 4–6}$$

$$B_S^s = \frac{1}{|S|}\sum_{m\in S} \varphi_s(F_m^s) \quad \text{Eq. 4}$$

$$R_S^s = \frac{1}{|S|}\sum_{m\in S} \hat{g}_m^s \odot \delta_s(F_m^s) \quad \text{Eq. 6}$$

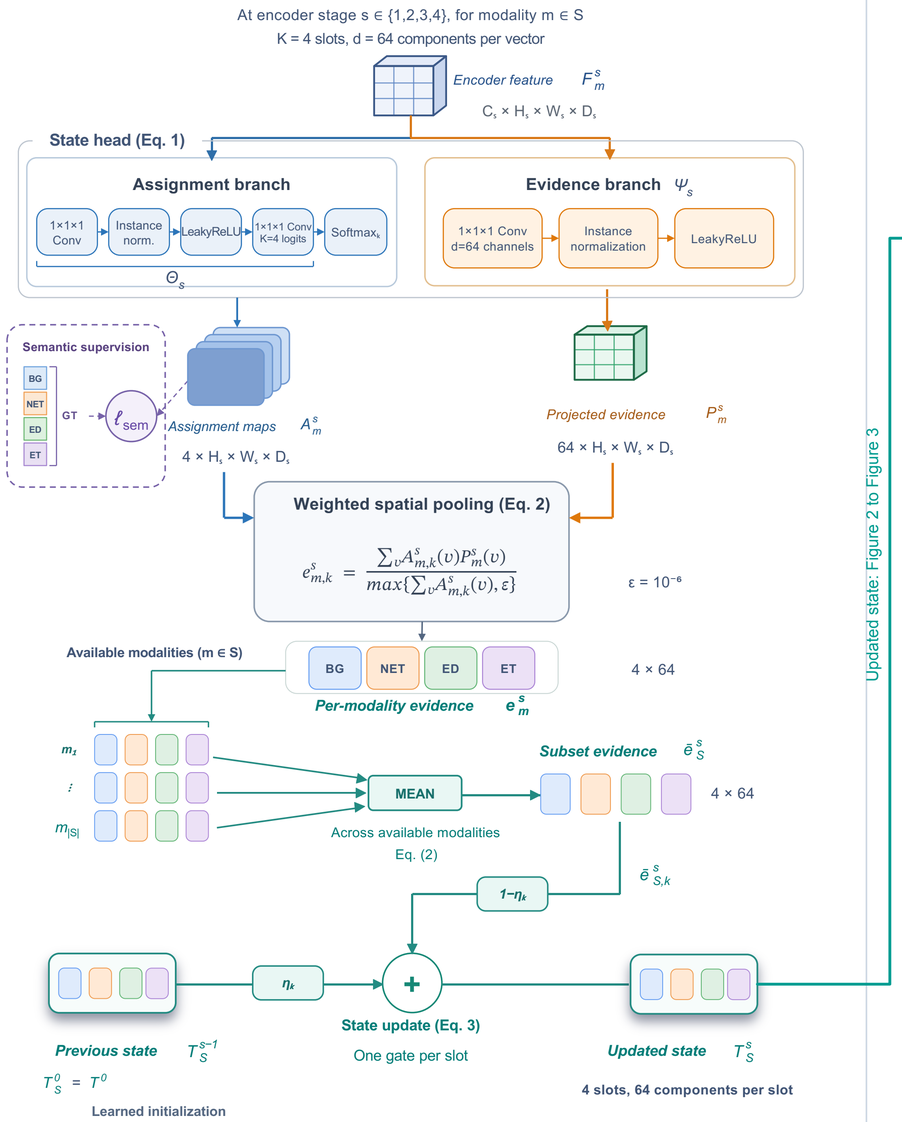
$$\hat{F}_S^s = \mathrm{IN}_s(B_S^s + \alpha_s\, R_S^s) \quad \text{Eq. 6}$$

Fused feature → shared decoder (all four scales)
Reset from learned T^0 at each forward pass / inference window.

Expand state construction and update

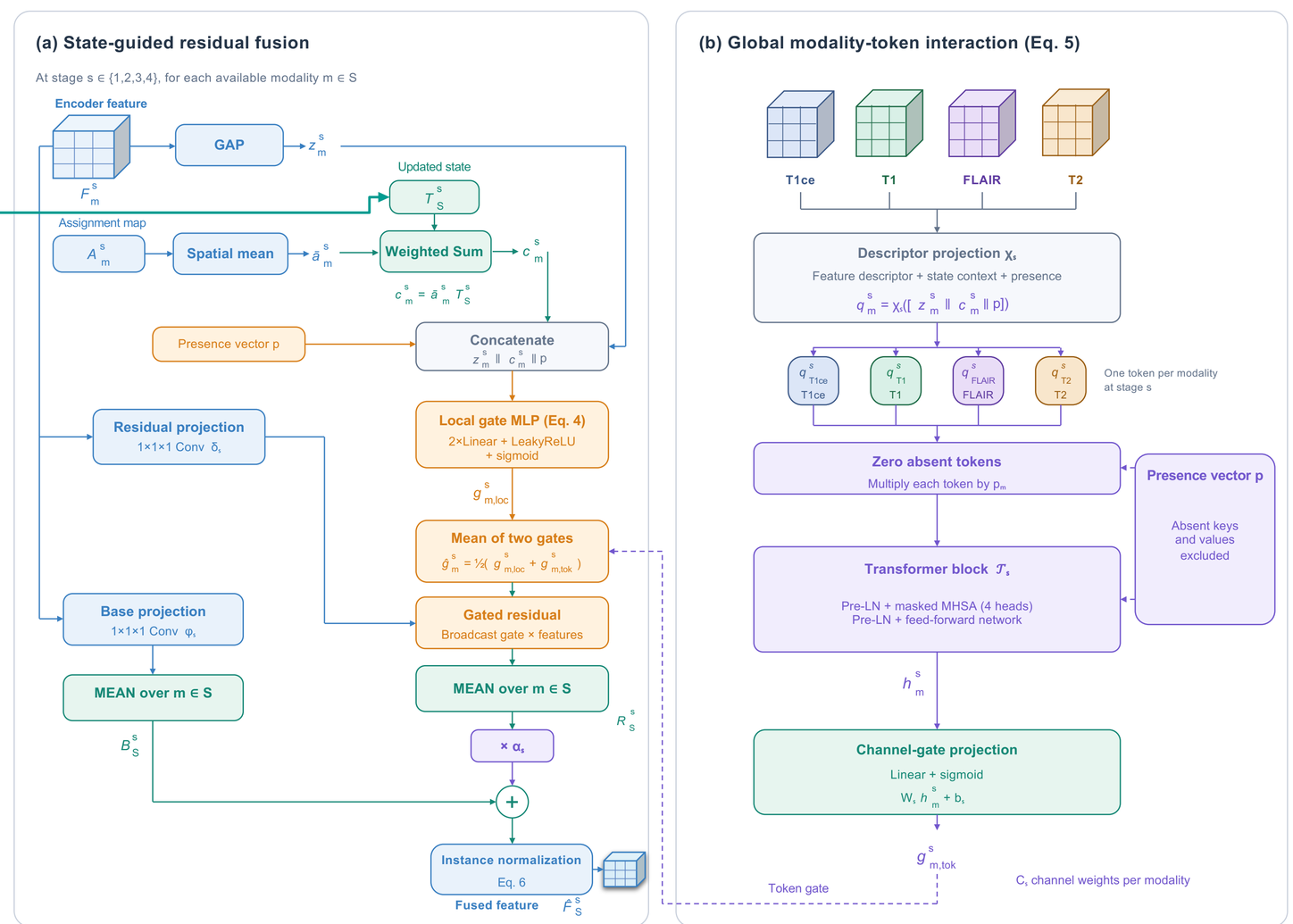
Expand fusion and token interaction

B Figure 2 · State update



Repeated at stages 1–4; four ordered state slots.

C Figure 3 · Fusion + token interaction



The teal connector carries the updated state from Figure 2 into the state-read operation in Figure 3.

Current features also feed both operations. Assignment maps A_m^s from Figure 2 supply the spatial-mean weights in Figure 3.

Blue: feature flow | Teal: tumor-state flow | Dashed arrows: expansion of the overview into its detailed modules